

Chapter 07 Homework — The Galois correspondence and cyclotomic fields (Problems 13–19)

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Standing assumptions. From here on we assume the fundamental theorem: for a finite Galois extension K/F , the maps $H \mapsto K^H$ and $L \mapsto \text{Gal}(K/L)$ are mutually inverse, inclusion-reversing bijections between the subgroups of $\text{Gal}(K/F)$ and the intermediate fields of K/F , with $[K : K^H] = |H|$; and L/F is Galois exactly when $\text{Gal}(K/L)$ is normal in $\text{Gal}(K/F)$, in which case $\text{Gal}(L/F) \cong \text{Gal}(K/F)/\text{Gal}(K/L)$. As before, $\zeta_n = e^{2\pi i/n}$, and all fields are subfields of $\mathbb{C}$ unless stated otherwise.

Problem 13. *The subfields of $\mathbb{Q}(\zeta_p)$*

Let p be an odd prime, $\zeta = \zeta_p$, $K = \mathbb{Q}(\zeta)$ and $G = \text{Gal}(K/\mathbb{Q})$.

- (a) Show that G is cyclic of order $p-1$.
- (b) Show that for each divisor d of $p-1$ there is exactly one intermediate field K_d with $[K_d : \mathbb{Q}] = d$, and that every such K_d is Galois over $\mathbb{Q}$ with cyclic Galois group.
- (c) Take $p = 13$. Check that 2 is a primitive root modulo 13, and use it to list every subgroup of $(\mathbb{Z}/13\mathbb{Z})^\times$ together with the degree over $\mathbb{Q}$ of the corresponding fixed field.
- (d) Let $H \leq G$ be the subgroup of index d , viewed inside $(\mathbb{Z}/p\mathbb{Z})^\times$, and set $\eta_H = \sum_{a \in H} \zeta^a$ (a Gaussian period). Show that $K_d = \mathbb{Q}(\eta_H)$. Hint: $\zeta, \zeta^2, \dots, \zeta^{p-1}$ is a $\mathbb{Q}$ -basis of K .

Problem 14. *The lattice for $\mathbb{Q}(\zeta_7)$*

Let $\zeta = \zeta_7$ and $K = \mathbb{Q}(\zeta)$.

- (a) Show that 3 is a primitive root modulo 7, and deduce that $\text{Gal}(K/\mathbb{Q}) = \langle \sigma \rangle$ where $\sigma(\zeta) = \zeta^3$.
- (b) Let $\alpha = \zeta + \zeta^2 + \zeta^4$ and $\beta = \zeta^3 + \zeta^5 + \zeta^6$. Show that $\alpha + \beta = -1$ and $\alpha\beta = 2$, and deduce that the unique quadratic subfield of K is $\mathbb{Q}(\sqrt{-7})$.
- (c) Let $\eta = \zeta + \zeta^{-1} = 2\cos(2\pi/7)$. Show that the minimal polynomial of η over $\mathbb{Q}$ is $x^3 + x^2 - 2x - 1$, and that $\mathbb{Q}(\eta)$ is the unique cubic subfield of K .
- (d) Draw the lattice of intermediate fields of $K/\mathbb{Q}$, and beside it the lattice of subgroups of $\text{Gal}(K/\mathbb{Q})$ drawn upside down (trivial subgroup at the top). Label every edge with the degree or index it represents, and match each subgroup with its fixed field.

Problem 15. *The real subfield of a cyclotomic field*

Let $n \geq 3$, $\zeta = \zeta_n$, $K = \mathbb{Q}(\zeta)$, and $K^+ = \mathbb{Q}(\zeta + \zeta^{-1})$.

- (a) Show that complex conjugation restricts to an automorphism c of K with $c(\zeta) = \zeta^{-1}$, and that c has order 2.
- (b) Show that ζ is a root of $x^2 - (\zeta + \zeta^{-1})x + 1 \in K^+[x]$, and deduce that $K^+ = K^{(c)} = K \cap \mathbb{R}$ and $[K : K^+] = 2$.
- (c) Deduce that $[K^+ : \mathbb{Q}] = \varphi(n)/2$ and that $K^+ = \mathbb{Q}(\cos(2\pi/n))$.
- (d) Take $n = 5$. Show that $K^+ = \mathbb{Q}(\sqrt{5})$, and deduce the exact value $\cos(2\pi/5) = \frac{1}{4}(\sqrt{5} - 1)$.

Problem 16. *Being Galois is not transitive*

Let $\alpha = \sqrt[4]{2} \in \mathbb{R}$.

- (a) Show that $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 4$ and that $\mathbb{Q}(\sqrt{2})$ is an intermediate field.
- (b) Show that $\mathbb{Q}(\sqrt{2})/\mathbb{Q}$ and $\mathbb{Q}(\alpha)/\mathbb{Q}(\sqrt{2})$ are both Galois.
- (c) Show that $\mathbb{Q}(\alpha)/\mathbb{Q}$ is not Galois, in two ways: by computing $\text{Aut}(\mathbb{Q}(\alpha)/\mathbb{Q})$, and by exhibiting an irreducible polynomial over $\mathbb{Q}$ with a root in $\mathbb{Q}(\alpha)$ that does not split there.
- (d) Find the smallest extension of $\mathbb{Q}(\alpha)$ that is Galois over $\mathbb{Q}$, and compute its degree.

Problem 17. *The Galois group of $x^4 - 2$*

Let $\alpha = \sqrt[4]{2}$ and $E = \mathbb{Q}(\alpha, i)$, so $[E : \mathbb{Q}] = 8$ by Problem 16.

- (a) Show that there is a $\sigma \in \text{Gal}(E/\mathbb{Q})$ with $\sigma(\alpha) = i\alpha$ and $\sigma(i) = i$, and let τ be complex conjugation. Show that σ has order 4, that τ has order 2, and that $\tau\sigma\tau^{-1} = \sigma^{-1}$. Conclude $\text{Gal}(E/\mathbb{Q}) \cong D_4$, the symmetries of a square.
- (b) The three subgroups of order 4 are $\langle \sigma \rangle$, $\langle \sigma^2, \tau \rangle$ and $\langle \sigma^2, \sigma\tau \rangle$. Show their fixed fields are $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{-2})$ respectively.
- (c) The five subgroups of order 2 are $\langle \sigma^2 \rangle$, $\langle \tau \rangle$, $\langle \sigma^2\tau \rangle$, $\langle \sigma\tau \rangle$ and $\langle \sigma^3\tau \rangle$. Show that their fixed fields are $\mathbb{Q}(i, \sqrt{2})$, $\mathbb{Q}(\alpha)$, $\mathbb{Q}(i\alpha)$, $\mathbb{Q}((1+i)\alpha)$ and $\mathbb{Q}((1-i)\alpha)$ respectively. Hint: it is enough to check that each listed field is fixed by the subgroup and has degree 4 over $\mathbb{Q}$; for the last two, look at the images under $\langle \sigma \rangle$.
- (d) Draw the lattice of all ten intermediate fields of $E/\mathbb{Q}$. Which of them are Galois over $\mathbb{Q}$? Reconcile your answer with Problem 16.

Problem 18. *Gauss sums and the quadratic subfield of $\mathbb{Q}(\zeta_p)$*

Let p be an odd prime, $\zeta = \zeta_p$, and write $\left(\frac{a}{p}\right)$ for the Legendre symbol (+1 if a is a nonzero square mod p , -1 if not). Define the Gauss sum

$$g = \sum_{k=1}^{p-1} \left(\frac{k}{p}\right) \zeta^k.$$

- (a) Show that $\mathbb{Q}(\zeta_p)$ has exactly one quadratic subfield.
- (b) Show that $\sigma_a(g) = \left(\frac{a}{p}\right) g$ for every $a \in (\mathbb{Z}/p\mathbb{Z})^\times$, where $\sigma_a(\zeta) = \zeta^a$.

- (c) Show that $g^2 = \left(\frac{-1}{p}\right)p = (-1)^{(p-1)/2}p$. Hint: write $l = km$ in the double sum and sum over k first.
- (d) Conclude that the quadratic subfield of $\mathbb{Q}(\zeta_p)$ is $\mathbb{Q}(\sqrt{p^*})$ with $p^* = (-1)^{(p-1)/2}p$. Check this against Problem 14 for $p = 7$, and compute it for $p = 5$ and $p = 13$.

Problem 19. ★ Every quadratic field is cyclotomic

Every quadratic field is $\mathbb{Q}(\sqrt{m})$ for a unique squarefree integer $m \neq 0, 1$; fix such an m .

- (a) Show that $\mathbb{Q}(\zeta_a) \subseteq \mathbb{Q}(\zeta_b)$ whenever $a \mid b$.
- (b) Show that $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{-2})$ all lie inside $\mathbb{Q}(\zeta_8)$.
- (c) Show that $\mathbb{Q}(\sqrt{m}) \subseteq \mathbb{Q}(\zeta_N)$ for $N = 4|m|$.
- (d) For $m = 3$, follow your proof of (c) to write $\sqrt{3}$ as a polynomial in ζ_{12} , and check that the result agrees with $2 \cos(\pi/6) = \zeta_{12} + \zeta_{12}^{-1}$.